

Kladáme poly. stupně ≤ 4

$$p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

ka podmínky $p(0)=1, p(1)=2, p(2)=-3$

a min $a_0^2 + a_1^2 + a_2^2 + a_3^2 + a_4^2$

doradíme $p(0)=a_0=1$

$$p(1)=a_4+a_3+a_2+a_1+a_0=2$$

$$p(2)=16a_4+8a_3+4a_2+2a_1+a_0=-4$$

$$a_0=1$$

$$a_4+a_3+a_2+a_1=1$$

$$16a_4+8a_3+4a_2+2a_1=-4$$

$$A^T(A\hat{x}-r)=0 \Leftrightarrow \text{derivován } A^T A \hat{x} = A^T r$$

Matice $A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \quad r = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$

$$A^T = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 5 & 9 & 17 \\ 3 & 5 & 9 & 17 & 33 \\ 5 & 9 & 17 & 33 & 65 \\ 9 & 17 & 33 & 65 & 129 \\ 17 & 33 & 65 & 129 & 257 \end{pmatrix}$$

$$A^T r = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \\ -10 \\ -22 \\ -46 \end{pmatrix}$$

Různice mířím

$$A^T A \hat{x} = A^T r \rightarrow \text{chceme nejmenší normu} \rightarrow \text{přičině}$$

$$\text{Řeší } \text{Ind}(A^T) \hat{x} = A^T r$$

$$A^T A (A^T)^{\dagger} A^T r$$

$$A^T (A A^T)^{\dagger} A^T r = A^T r$$

$$(A A^T)^{\dagger} A^T r = r$$

$$A A^T = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 31 \\ 1 & 31 & 341 \end{pmatrix}$$

$$(A A^T)^{\dagger} = I$$

$$y = \begin{pmatrix} 1/5 \\ 1 \\ -1/5 \end{pmatrix}$$

$$\hat{x} = A^T y = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \\ 0 & 1 & 8 \\ 0 & 1 & 16 \end{pmatrix} \begin{pmatrix} 1/5 \\ 1 \\ -1/5 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 \\ 4/5 \\ 3/5 \\ 1/5 \\ -3/5 \end{pmatrix}$$

$$p(x) = -\frac{3}{5}x^4 + \frac{1}{5}x^3 + \frac{3}{5}x^2 + \frac{4}{5}x + 1$$